

VORONEZH CHERTOVITSKY, Russian Federation

WMO: 341220

Lat: 51.814N

Lon: 39.230E

Elev: 157

StdP: 99.46

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.2	-21.0	-27.2	0.3	-24.1	-23.9	0.4	-20.9	12.0	-1.4	10.7	-3.0	2.3	310	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	33.0	19.1	30.9	18.6	28.9	18.1	20.8	27.8	20.0	27.1	19.3	26.2	4.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.9	23.2	17.9	13.1	22.3	17.0	12.4	21.7	60.8	27.7	58.1	27.2	55.7	25.8	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.3	7.3	-27.8	35.6	4.1	2.1	-30.8	37.1	-33.1	38.3	-35.4	39.4	-38.4	40.9		
(5)			-27.3	22.8	4.0	1.4	-30.3	23.8	-32.6	24.7	-34.9	25.4	-37.8	26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-7.1	-6.7	-1.1	8.1	15.3	18.2	20.7	20.2	13.8	7.2	1.0	-3.2	(6)
(7)		DBStd	11.11	7.04	6.80	5.26	4.53	4.41	3.88	3.36	3.98	3.77	4.59	4.57	5.30	(7)
(8)		HDD10.0	2239	531	466	344	90	8	1	0	0	9	112	270	408	(8)
(9)		HDD18.3	4274	789	699	603	308	113	50	13	25	142	346	519	666	(9)
(10)		CDD10.0	1248	0	0	1	32	173	247	333	314	124	24	1	0	(10)
(11)		CDD18.3	241	0	0	0	0	20	46	88	81	6	0	0	0	(11)
(12)		CDH23.3	2737	0	0	0	9	290	497	893	947	101	0	0	0	(12)
(13)		CDH26.7	982	0	0	0	1	74	152	335	401	19	0	0	0	(13)
(14)	Wind	WSAvg	3.6	3.9	4.2	4.1	3.8	3.4	3.0	3.0	3.2	3.6	3.9	4.2	(14)	
(15)	Precipitation	PrecAvg	566	39	36	34	40	47	64	58	57	50	44	44	(15)	
(16)		PrecMax	801	83	66	64	148	106	119	114	184	174	126	109	112	(16)
(17)		PrecMin	403	16	6	1	4	5	18	6	4	5	9	3	14	(17)
(18)		PrecStd	103	16	15	17	30	26	27	27	36	44	29	30	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	5.1	14.9	24.9	31.1	33.1	35.8	37.2	29.1	21.1	13.9	8.1	(19)
(20)			MCWB	4.3	4.0	8.0	12.9	18.5	18.7	18.9	18.7	15.7	13.2	11.1	7.5	(20)
(21)		2%	DB	3.0	3.2	10.9	21.2	28.8	30.2	32.2	33.8	26.2	18.2	11.2	6.1	(21)
(22)			MCWB	2.5	2.3	6.0	11.6	16.9	18.2	19.4	18.9	15.1	12.8	9.3	5.7	(22)
(23)		5%	DB	1.9	2.1	7.9	19.0	26.8	28.1	30.1	31.0	23.8	16.1	8.9	4.8	(23)
(24)			MCWB	1.5	1.3	4.9	10.9	16.1	18.5	19.1	18.7	15.0	12.2	7.5	4.3	(24)
(25)		10%	DB	0.9	1.2	5.8	16.2	24.2	26.1	28.0	28.2	21.1	14.0	7.0	3.0	(25)
(26)			MCWB	0.6	0.6	3.5	9.2	15.2	17.6	18.3	17.8	14.0	10.7	6.2	2.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	4.2	8.8	14.2	19.9	21.7	21.8	21.6	18.3	15.6	11.8	7.5	(27)
(28)			MCDWB	4.5	4.5	13.8	21.9	28.7	27.0	29.5	28.9	22.6	18.6	13.5	7.9	(28)
(29)		2%	WB	2.4	2.5	6.5	12.7	18.1	20.4	20.9	20.3	16.7	13.9	9.7	5.6	(29)
(30)			MCDWB	2.7	2.9	9.4	18.7	25.5	26.3	28.0	28.4	22.2	16.9	11.2	5.9	(30)
(31)		5%	WB	1.2	1.3	5.0	11.6	17.0	19.5	20.1	19.5	15.6	12.6	7.5	4.0	(31)
(32)			MCDWB	1.5	1.7	7.1	17.3	24.3	25.5	27.1	27.7	21.4	15.7	8.4	4.4	(32)
(33)		10%	WB	0.4	0.6	3.5	10.3	16.0	18.5	19.4	18.6	14.7	11.3	6.1	2.4	(33)
(34)			MCDWB	0.6	1.1	5.3	15.0	22.1	24.2	25.8	26.4	20.4	13.9	6.6	2.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.6	6.5	7.8	10.8	12.3	11.5	11.7	12.8	11.1	8.0	5.1	4.6	(35)
(36)			MCDBR	3.8	4.4	10.7	14.4	15.6	14.3	14.6	16.1	15.3	11.5	7.9	4.3	(36)
(37)			MCWBR	3.6	4.0	6.7	7.3	6.4	5.8	5.1	5.4	6.8	6.8	6.4	4.2	(37)
(38)		5% WB	MCDBR	3.7	4.2	9.3	12.7	13.3	12.1	12.0	13.4	12.4	9.8	6.2	4.1	(38)
(39)			MCWBR	3.7	4.0	6.2	7.1	6.5	5.8	5.0	5.8	6.5	6.6	5.6	4.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.304	0.340	0.364	0.440	0.396	0.389	0.420	0.436	0.388	0.361	0.338	0.306	(40)	
(41)		taud	2.124	2.028	2.098	2.090	2.317	2.354	2.276	2.216	2.334	2.353	2.276	2.175	(41)	
(42)		Ebn at Noon	667	743	813	792	857	866	829	790	790	733	634	598	(42)	
(43)		Edh at Noon	75	111	126	144	121	117	125	126	100	81	66	62	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.95	1.98	3.36	4.18	5.72	5.95	5.83	5.04	3.33	1.75	0.77	0.54	(44)	
(45)		RadStd	0.22	0.26	0.54	0.37	0.46	0.53	0.48	0.43	0.44	0.25	0.16	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	-193	+124	N/A

Nomenclature: See separate page